

Tutorial Problems: Separation of Variables, Similarity Solutions, Complex Variables

[For tutorial classes in the week commencing 19th February 2024]

Separation of Variables, Similarity Solutions

1. Suppose that $u(x, t)$ represents the temperature at time $t > 0$ in a 1-dimensional wire occupying the spatial domain $0 \leq x \leq \pi$. Solve the heat equation

$$u_t = \kappa u_{xx},$$

where $\kappa > 0$ is the constant of thermal diffusivity, subject to the boundary conditions

$$u(0, t) = u(\pi, t) = 0, \quad t > 0,$$

and for three different choices of initial condition on the domain $0 \leq x \leq \pi$:

- (i) For $u(x, 0) = 2 \sin x$;
(ii) For $u(x, 0) = \begin{cases} x & \text{for } 0 \leq x \leq \frac{\pi}{2} \\ (\pi - x) & \text{for } \frac{\pi}{2} \leq x \leq \pi. \end{cases}$;
(iii) For $u(x, 0) = \sin^3 x$.

[Hint: For part (iii), which is easier than (ii), it may be helpful to recall that $\sin(3x) = 3 \sin x - 4 \sin^3 x$.]

$$u_t = \kappa u_{xx}$$

$$\text{Let } u(x, t) = X(x)T(t), \text{ then } X(x) \frac{dT(t)}{dt} = \kappa T(t) \frac{d^2 X(x)}{dx^2}$$

$$\therefore \frac{X''(x)}{X(x)} = \frac{T'(t)}{\kappa T(t)} = -\lambda^2$$

$$\therefore \begin{cases} X''(x) + \lambda^2 X(x) = 0 \\ T'(t) + \lambda^2 \kappa T(t) = 0 \end{cases}$$

$$\text{Let } X(x) = A \cos(\lambda x) + B \sin(\lambda x)$$

$$\therefore X''(x) = -\lambda^2 X(x)$$

$$\therefore u(0, t) = 0 \quad \therefore X(0) = 0 \quad \therefore A = 0$$

$$\therefore u(x, t) = 0 \quad \therefore X(x) = 0$$

$$\therefore B \sin(\pi \lambda) = 0 \quad \therefore \sin(\pi \lambda) = 0$$

$$\therefore \lambda = n \text{ where } n = 0, 1, 2, 3, \dots$$

$$\therefore X(x) = B \sin(n x)$$

$$\text{Let } T(t) = C e^{-\lambda^2 K t} \quad \therefore T'(t) = -\lambda^2 K T(t)$$

$$\therefore T(t) = C e^{-n^2 K t}$$

$$\therefore u(x, t) = B \sin(n x) C e^{-n^2 K t}$$

$$= \sum_{n=1}^{\infty} B_n \sin(n x) e^{-n^2 K t} \text{ where } B_n = B C$$

$$\textcircled{1} \text{ for } u(x, 0) = 2 \sin x$$

$$u(x, 0) = \sum_{n=1}^{\infty} B_n \sin(n x) = 2 \sin x$$

$$\begin{aligned} \therefore B_n &= \frac{2}{\pi} \int_0^{\pi} 2 \sin x \sin(n x) dx \quad \frac{1}{2j} (e^{jx} - e^{-jx}) \quad \frac{1}{2j} (e^{jn x} - e^{-jn x}) \\ &= \frac{2}{\pi} \left[\frac{e^{j(n+1)x}}{j(n+1)} - \frac{e^{j(n-1)x}}{j(n-1)} - \frac{e^{-j(n+1)x}}{j(n+1)} + \frac{e^{-j(n-1)x}}{j(n-1)} \right]_0^{\pi} \\ &= \frac{2}{\pi} \left[2 \left[-\sin(n x) \cos x + n \cos(n x) \sin x \right]_0^{\pi} \right] \\ &\quad + n^2 \int_0^{\pi} 2 \sin x \sin(n x) dx \end{aligned}$$

$$\therefore B_n = \frac{4}{\pi(n^2-1)} \left[\sin(n x) \cos x - n \cos(n x) \sin x \right]_0^{\pi}$$

$$= \frac{4}{\pi(n^2-1)} [-\sin(n \pi)] = 0 = 2 \text{ (only for } n=1)$$

$$\textcircled{2} \text{ for } u(x, 0) = \begin{cases} x & \text{for } 0 \leq x \leq \frac{\pi}{2} \\ \pi - x & \text{for } \frac{\pi}{2} \leq x \leq \pi \end{cases}$$

$$u(x, 0) = \sum_{n=1}^{\infty} B_n \sin(nx)$$

$$\begin{aligned} \therefore B_n &= \frac{2}{\pi} \left(\int_0^{\frac{\pi}{2}} x \sin(nx) dx + \int_{\frac{\pi}{2}}^{\pi} (\pi - x) \sin(nx) dx \right) \\ &= \frac{2}{\pi} \left(\left[\frac{1}{n^2} \sin(nx) - \frac{x}{n} \cos(nx) \right]_0^{\frac{\pi}{2}} + \left[\frac{x-\pi}{n} \cos(nx) - \frac{1}{n^2} \sin(nx) \right]_{\frac{\pi}{2}}^{\pi} \right) \\ &= \frac{2}{\pi} \left[\frac{1}{n^2} \sin\left(\frac{\pi}{2}n\right) - \frac{\pi}{2n} \cos\left(\frac{\pi}{2}n\right) + \cancel{\frac{\pi}{2n}} - \frac{1}{n^2} \sin(n\pi) \right. \\ &\quad \left. + \frac{\pi}{2n} \cos\left(\frac{\pi}{2}n\right) + \frac{1}{n^2} \sin\left(\frac{\pi}{2}n\right) \right] \\ &= \frac{2}{\pi} \left[\frac{2}{n^2} \sin\left(\frac{\pi}{2}n\right) + \cancel{\frac{\pi}{2n}} \right] \end{aligned}$$

$$\therefore B_n = \begin{cases} \cancel{\frac{1}{n^2}} & n \text{ is odd} \\ \frac{2}{n^2} + \cancel{\frac{1}{n}} & n = 1 + 2n \text{ (n is even)} \\ -\frac{2}{n^2} + \cancel{\frac{1}{n}} & n = 3 + 2n \text{ (n is even)} \end{cases}$$

see solution

$$\textcircled{3} \text{ for } u(x, 0) = \sin^3(x) = \frac{3}{4} \sin x - \frac{1}{4} \sin(3x)$$

$$\begin{aligned} \therefore B_n &= \frac{2}{\pi} \int_0^{\pi} \frac{3}{4} \sin x \sin(nx) dx - \frac{2}{\pi} \int_0^{\pi} \frac{1}{4} \sin(3x) \sin(nx) dx \\ &\quad \begin{cases} 2x & n \neq 1 \\ 0 & n \text{ otherwise} \end{cases} \quad \begin{cases} 2x & n \neq 3 \\ 0 & n \text{ otherwise} \end{cases} \\ &= \frac{3}{2\pi} \int_0^{\pi} \sin x \sin(nx) dx - \frac{1}{2\pi} \int_0^{\pi} \sin(3x) \sin(nx) dx \\ &= 0 - \frac{1}{2\pi} \frac{9}{9-n^2} \left[-\frac{1}{3} \sin(nx) \cos(3x) + \frac{1}{9} n \sin(3x) \cos(nx) \right]_0^{\pi} \\ &= \frac{9}{2(n^2-9)\pi} \left[\frac{1}{3} \sin(nx) + 0 + 0 - 0 \right] = 0 \end{aligned}$$

see solution

2. Consider a rectangular plate occupying the space $0 \leq x \leq a$ and $0 \leq y \leq b$. Let $T(x, y)$ denote the temperature of the plate at position (x, y) . The three sides $x = 0$, $x = a$ and $y = 0$ are maintained at zero temperature $T = 0$. The fourth side $y = b$ is maintained at temperature $T = 1$.

The heat distribution satisfies Laplace's equation $T_{xx} + T_{yy} = 0$ in the rectangle. Assuming a separation of variables solution of the form $T(x, y) = X(x)Y(y)$, show that these functions satisfy

$$X_n(x) = A_n \sin\left(\frac{n\pi x}{a}\right), \quad Y_n(y) = B_n \left(e^{\frac{n\pi y}{a}} - e^{-\frac{n\pi y}{a}}\right), \quad n \geq 1,$$

where A_n, B_n are constants. Hence show that the solution is

$$T(x, y) = \frac{4}{\pi} \sum_{n=0}^{\infty} \frac{1}{2n+1} \sin\left(\frac{(2n+1)\pi x}{a}\right) \left[\frac{\sinh\left(\frac{(2n+1)\pi y}{a}\right)}{\sinh\left(\frac{(2n+1)\pi b}{a}\right)} \right].$$

$$T_{xx} + T_{yy} = 0 \quad \therefore X''(x)Y(y) + X(x)Y''(y) = 0$$

$$\therefore \frac{X''(x)}{X(x)} = -\frac{Y''(y)}{Y(y)} = -\lambda^2$$

$$\therefore \begin{cases} X''(x) + \lambda^2 X(x) = 0 \\ Y''(y) - \lambda^2 Y(y) = 0 \end{cases}$$

$$\text{Let } X(x) = A \cos(\lambda x) + B \sin(\lambda x)$$

$$\therefore X(0) = A = 0$$

$$\therefore X(a) = B \sin(\lambda a) = 0 \quad \therefore \lambda = \frac{n\pi}{a}, \quad n = 1, 2, 3, \dots$$

$$\therefore X(x) = A_n \sin\left(\frac{n\pi x}{a}\right)$$

$$\text{Let } Y(y) = C e^{\lambda y} + D e^{-\lambda y}$$

$$\therefore Y(0) = C + D = 0 \quad \therefore D = -C$$

$$\therefore Y(b) = B_n \left(e^{\frac{n\pi}{a}b} - e^{-\frac{n\pi}{a}b}\right), \quad n = 1, 2, 3, \dots$$

$$\therefore T(x, y) = \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi x}{a}\right) B_n \left(e^{\frac{n\pi}{a}y} - e^{-\frac{n\pi}{a}y}\right)$$

$$\therefore T(x, b) = \sum_{n=1}^{\infty} A_n B_n \sin\left(\frac{n\pi x}{a}\right) \left[e^{\frac{n\pi}{a}b} - e^{-\frac{n\pi}{a}b}\right] = 1$$

$$\therefore A_n B_n = \frac{2}{a} \int_0^a \sin\left(\frac{n\pi x}{a}\right) 2 \sinh\left(\frac{n\pi b}{a}\right) dx$$

$$= \frac{2}{a} 2 \sinh\left(\frac{n\pi b}{a}\right) \int_0^a \sin\left(\frac{n\pi x}{a}\right) dx$$

$$= \frac{2}{a} 2 \sinh\left(\frac{n\pi b}{a}\right) \frac{a}{n\pi} \left[-\cos\left(\frac{n\pi x}{a}\right) \right]_0^a$$

$$= \frac{2}{n\pi} 2 \sinh\left(\frac{n\pi b}{a}\right) \left[-\cos(n\pi) + 1 \right]$$

$$= \begin{cases} 0 & n \text{ is even} \\ \frac{4}{n\pi} \left[2 \sinh\left(\frac{n\pi b}{a}\right) \right] & n \text{ is odd} \end{cases}$$

$$\therefore \text{Let } n = 2n+1$$

$$\therefore A_n B_n = \frac{4}{(2n+1)\pi} 2 \sinh\left[\frac{(2n+1)\pi b}{a}\right]$$

$$\therefore T(x, y) = \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{1}{2n+1} \sin\left[\frac{(2n+1)\pi x}{a}\right] \frac{\sinh\left[\frac{(2n+1)\pi y}{a}\right]}{\sinh\left[\frac{(2n+1)\pi b}{a}\right]}$$

3. The three-dimensional diffusion equation with spherical symmetry is given by

$$u_t = \kappa \left(u_{rr} + \frac{2}{r} u_r \right).$$

Using separation of variables in the form $u(r, t) = R(r)T(t)$ and then changing the variable R to $R(r) = r^{-1}S(r)$, show that

$$\frac{T'(t)}{\kappa T(t)} = \frac{S''(r)}{S(r)}.$$

Hence, show that if u is finite at every point within a sphere $r = a$ and is subject to the condition $u = 0$ on the sphere's surface $r = a$ for all $t > 0$ then the solution of this PDE within the sphere for all $t > 0$ takes the form

$$u(r, t) = \sum_{n=1}^{\infty} B_n \frac{\sin\left(\frac{n\pi r}{a}\right)}{r} e^{-\frac{\kappa n^2 \pi^2 t}{a^2}}.$$

$$R(r) T'(t) = \kappa \left[R''(r) + \frac{2}{r} R'(r) \right] T(t)$$

$$\therefore R(r) = \frac{1}{r} S(r) \quad \therefore R'(r) = \frac{1}{r} S'(r) - \frac{1}{r^2} S(r)$$

$$\therefore R''(r) = \frac{1}{r} S''(r) - \frac{2}{r^2} S'(r) + \frac{2}{r^3} S(r)$$

$$\therefore R''(r) + \frac{2}{r} R'(r) = \frac{1}{r} S''(r)$$

$$\therefore S(r) T'(t) = \kappa S''(r) T(t)$$

$$\therefore \frac{S''(r)}{S(r)} = \frac{T'(t)}{\kappa T(t)} = -\lambda^2$$

$$\therefore \text{Let } S(r) = A \sin(\lambda r) + B \cos(\lambda r)$$

$$\text{and } T(t) = C e^{-\lambda^2 \kappa t}$$

$$\therefore S(0) = B = 0$$

$$\therefore S(a) = A \sin(\lambda a) = 0$$

$$\therefore \lambda a = n\pi \text{ for } n = 1, 2, 3, \dots$$

$$\therefore \lambda = \frac{n\pi}{a}$$

Therefore, $S(r) = A \sin\left(\frac{n\pi r}{a}\right)$,

and $T(t) = Ce^{-\left(\frac{n\pi}{a}\right)^2 kt}$

$$\therefore u(r, t) = R(r) T(t)$$

$$= \frac{1}{r} S(r) T(t)$$

$$= \sum_{n=1}^{\infty} B_n \frac{\sin\left(\frac{n\pi r}{a}\right)}{r} e^{-\left(\frac{n\pi}{a}\right)^2 kt}$$

4. Show that the PDE

$$u_t = u_{xx} - \frac{1}{2\sqrt{t}}u_x$$

has a similarity solution of the form

$$u(x, t) = t^m f(\eta), \quad \eta = \frac{x}{\sqrt{t}}.$$

Derive the ODE that only involves $f(\eta)$, its derivatives, η and the constant m .

Verify that for $m = -\frac{1}{2}$ a solution is given by

$$f(\eta) = \frac{1}{2}e^{-\frac{1}{4}(\eta-1)^2}$$

and write out the corresponding solution $u(x, t)$ to the original PDE. For this $u(x, t)$, also compute the limits

$$\lim_{t \rightarrow \infty} u(x, t) \quad \text{and} \quad \lim_{t \rightarrow 0^+} u(x, t)$$

for all real values of x .

$$\begin{aligned} u_t &= m t^{m-1} f(\eta) - \frac{1}{2t} \frac{x}{\sqrt{t}} t^m f'(\eta) \\ &= t^{m-1} \left[m f(\eta) - \frac{1}{2} \eta f'(\eta) \right] \end{aligned}$$

$$u_x = t^{m-\frac{1}{2}} f'(\eta)$$

$$u_{xx} = t^{m-1} f''(\eta)$$

$$\therefore t^{m-1} \left[m f(\eta) - \frac{1}{2} \eta f'(\eta) \right] = t^{m-1} f''(\eta) - \frac{1}{2} t^{m-1} f'(\eta)$$

$$\therefore f''(\eta) + \frac{1}{2} (\eta - 1) f'(\eta) - m f(\eta) = 0$$

$$\text{Let } g = f(\eta) \quad \therefore \frac{d^2 g}{d\eta^2} + \frac{1}{2} (\eta - 1) \frac{dg}{d\eta} - m g = 0$$

$$\text{for } m = -\frac{1}{2}, \quad \frac{d^2 g}{dy^2} + \frac{1}{2}(y-1) \frac{dg}{dy} + \frac{1}{2}g = 0$$

$$\text{when } g = f(y) = \frac{1}{2} e^{-\frac{1}{4}(y-1)^2}$$

$$\frac{d^2 g}{dy^2} + \frac{1}{2}(y-1) \frac{dg}{dy} + \frac{1}{2}g$$

$$= \frac{1}{2} \left[-\frac{1}{2} + \frac{1}{4}(y-1)^2 - \frac{1}{4}(y-1)^2 + \frac{1}{2} \right] e^{-\frac{1}{4}(y-1)^2} = 0$$

$$\therefore u(x, t) = \frac{1}{2} t^{-\frac{1}{2}} e^{-\frac{1}{4}(y-1)^2}$$

$$\therefore \lim_{t \rightarrow \infty} u(x, t) = 0$$

$$\lim_{\substack{x \rightarrow \infty \\ t \rightarrow 0}} u(x, t) = 0^{\infty} \text{ for all real value } x$$

$$= 0 \text{ for } x = 0$$

Complex Variables

5. Verify that the following functions satisfy the Cauchy-Riemann equations:

(i) $u = x, v = y$

(ii) $u = e^x \cos y, v = e^x \sin y$

(iii) $u = x^3 - 3xy^2, v = 3x^2y - y^3$

i) $\frac{du}{dx} = 1, \frac{du}{dy} = 0, \frac{dv}{dx} = 0, \frac{dv}{dy} = 1$

$$\therefore \frac{du}{dx} = \frac{dv}{dy} \text{ and } \frac{dv}{dx} = -\frac{du}{dy}$$

ii) $\frac{du}{dx} = e^x \cos y, \frac{du}{dy} = -e^x \sin y,$

$$\frac{dv}{dx} = e^x \sin y, \frac{dv}{dy} = e^x \cos y$$

$$\therefore \frac{du}{dx} = \frac{dv}{dy} \text{ and } \frac{dv}{dx} = -\frac{du}{dy}$$

iii) $\frac{du}{dx} = 3x^2 - 3y^2, \frac{du}{dy} = -6xy,$

$$\frac{dv}{dx} = 6xy, \frac{dv}{dy} = 3x^2 - 3y^2$$

$$\therefore \frac{du}{dx} = \frac{dv}{dy} \text{ and } \frac{dv}{dx} = -\frac{du}{dy}$$

6. Show that the following functions are not analytic at any point:

(i) $f(z) = z|z|$

(ii) $f(z) = \bar{z}$

(i) $f(z) = (x+iy)|x+iy|$

$$= (x+iy)\sqrt{x^2+y^2}$$

$$= x\sqrt{x^2+y^2} + iy\sqrt{x^2+y^2}$$

$$\therefore u = x\sqrt{x^2+y^2}, \quad v = y\sqrt{x^2+y^2}$$

$\therefore u, v$ are linear

$$\frac{du}{dx} = \sqrt{x^2+y^2} + \frac{x^2}{\sqrt{x^2+y^2}}$$

$$\frac{dv}{dy} = \sqrt{x^2+y^2} + \frac{y^2}{\sqrt{x^2+y^2}}$$

$$\frac{dv}{dx} = \frac{xy}{\sqrt{x^2+y^2}}$$

$$\frac{du}{dy} = \frac{xy}{\sqrt{x^2+y^2}}$$

$$\therefore \frac{du}{dx} \neq \frac{dv}{dy} \text{ and } \frac{dv}{dx} \neq \frac{du}{dy}$$

$\therefore$ not analytic at any point

$$(ii) f(z) = \bar{z} = x - iy$$

$$u = x, v = -y$$

$\therefore u, v$ are linear

$$\frac{du}{dx} = 1, \frac{du}{dy} = 0, \frac{dv}{dx} = 0, \frac{dv}{dy} = -1$$

$$\therefore \frac{du}{dx} \neq \frac{dv}{dy}, \frac{dv}{dx} \neq -\frac{du}{dy}$$

$\therefore$ not analytic at any point

7. Show that the following functions $u(x, y)$ are harmonic and then use the Cauchy-Riemann equations to determine each function's conjugate function v . Find the analytic function $f(z) = u + iv$ in each case:

(i) $u(x, y) = x^2 + 2x - y^2$

(ii) $u(x, y) = xy$

(i) $u_{xx} = 2, u_{yy} = -2$

$\therefore u_{xx} + u_{yy} = 0 \quad \therefore \text{harmonic}$

$u_x = 2x + 2 = v_y \quad \therefore v = 2xy + 2y + A$

$-u_y = 2y = v_x \quad \therefore v = 2xy + B$

$\therefore v = 2xy + \textcolor{blue}{2y} + \textcolor{blue}{C}$

$\therefore f(z) = x^2 + 2x - y^2 + 2ixy + \textcolor{blue}{2iy} + \textcolor{blue}{iC}$
 $= \textcolor{blue}{z^2 + 2z + iC}$

(ii) $u_{xx} = u_{yy} = 0 \quad \therefore u_{xx} + u_{yy} = 0 \quad \therefore \text{harmonic}$

$\therefore u_x = v_y = y \quad \therefore v = \frac{1}{2}y^2 + A$

$-u_y = v_x = -x \quad v = -\frac{1}{2}x^2 + B$

$\therefore v = -\frac{1}{2}x^2 + \frac{1}{2}y^2 + C$

$\therefore f(z) = xy - \frac{1}{2}ix^2 + \frac{1}{2}iy^2 + iC$
 $= \textcolor{blue}{-\frac{1}{2}iz^2 + iC}$

8. Suppose a, b, c, d satisfy $ad - bc \neq 0$. At which points of the complex plane is the Möbius transformation

$$w = f(z) = \frac{az + b}{cz + d}$$

conformal?

$$w = f(z) = \frac{az + b}{cz + d} \quad \text{where } ad - bc \neq 0$$

① w analytic on subset G

$$w = \frac{a(x+iy) + b}{c(x+iy) + d} = \frac{(ax+b) + iay}{(cx+d) + icy}$$

$$= \frac{[(ax+b) + iay][(cx+d) - icy]}{(cx+d)^2 + (cy)^2}$$

$$\therefore u = \frac{(ax+b)(cx+d) + acy^2}{(cx+d)^2 + (cy)^2}$$

$$v = \frac{ay(cx+d) - cy(ax+b)}{(cx+d)^2 + (cy)^2}$$

$$\begin{aligned} \therefore u_x &= \frac{[a(cx+d) + c(ax+b)][(cx+d)^2 + (cy)^2] - [(ax+b)(cx+d) + acy^2] 2(cx+d)}{[(cx+d)^2 + (cy)^2]^2} \\ &= v_y \end{aligned}$$

similarly, $u_y = -v_x$

② $f'(z) \neq 0$ for any $z \in G$

$$\text{when } f'(z) = \frac{a(cz+d) - (az+b)c}{(cz+d)^2} = \frac{ad-bc}{(cz+d)^2} \neq 0$$

$\therefore ad - bc \neq 0$, conformal